# dataiku

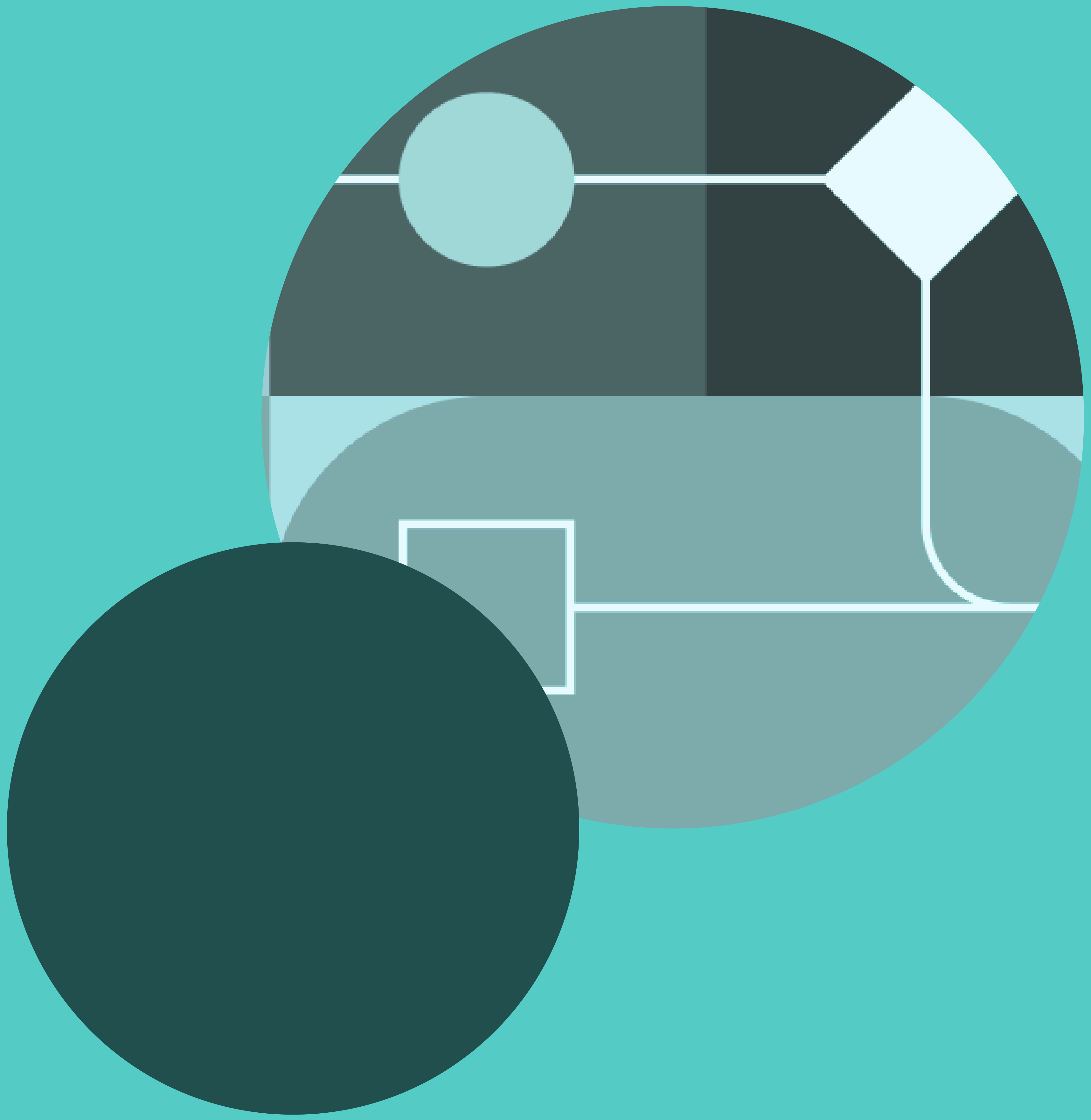
Prateek Rana

# Agenda

1. Introduction
2. Data understanding
3. Exploratory data analysis (EDA)
4. Feature engineering
5. Modeling and evaluation
6. Results
7. Recommendations
8. Limitations
9. Future work

# Introduction

- Project overview
- Question at hand: **“Who earns more than $50K?”**
- Key Goals:
  - Predict individual earns more or less than $50K
  - Most influential features
  - Build robust model
  - Evaluation of model
  - Actionable insights



# Data understanding

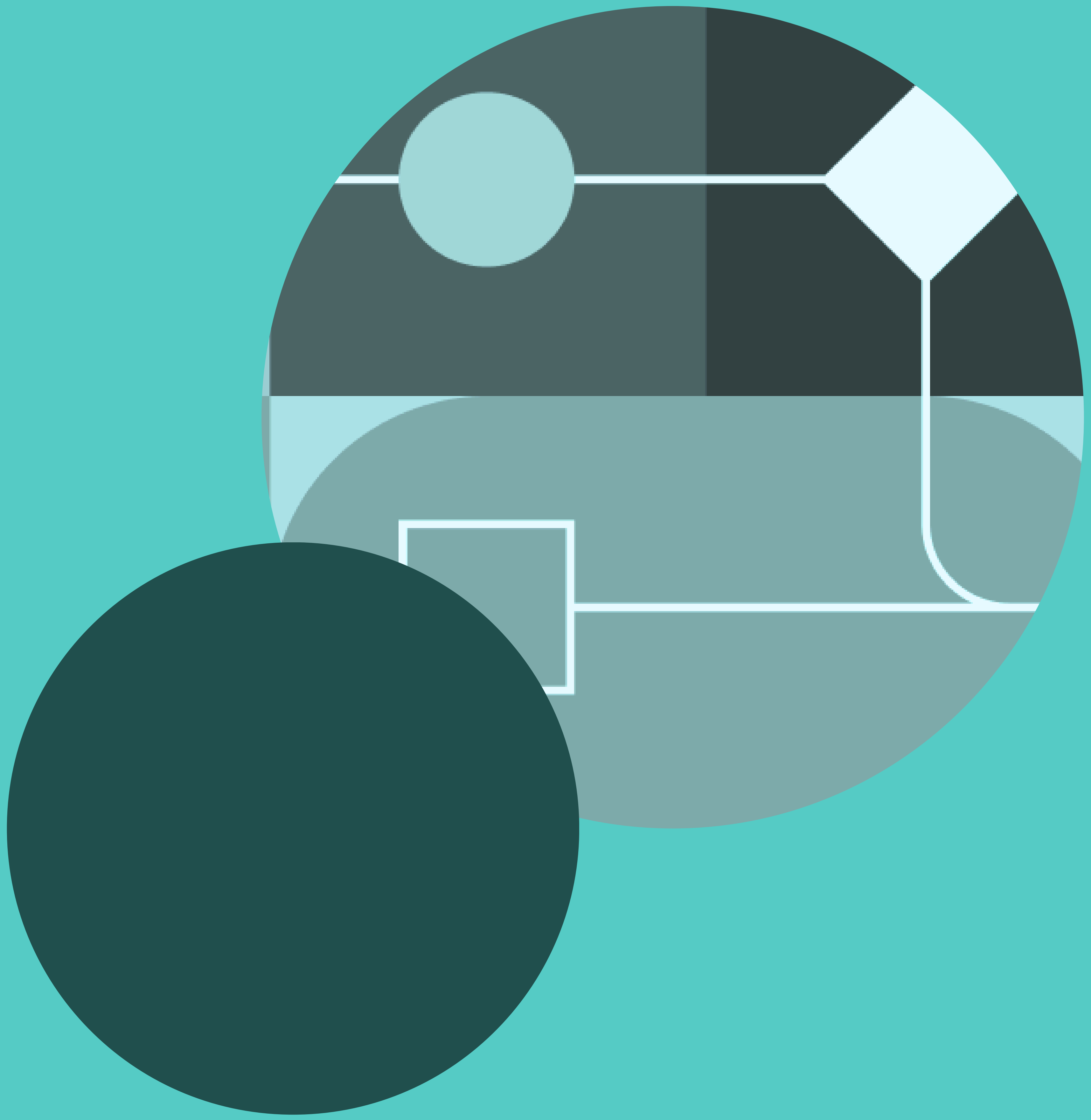
- Size: ~200,000
- Scope: Demographic, employment and financial variables
- Key variables:
  - instance_weight (ignored while modeling)
  - income_level
  - age, sex
  - class_of_worker

## Source

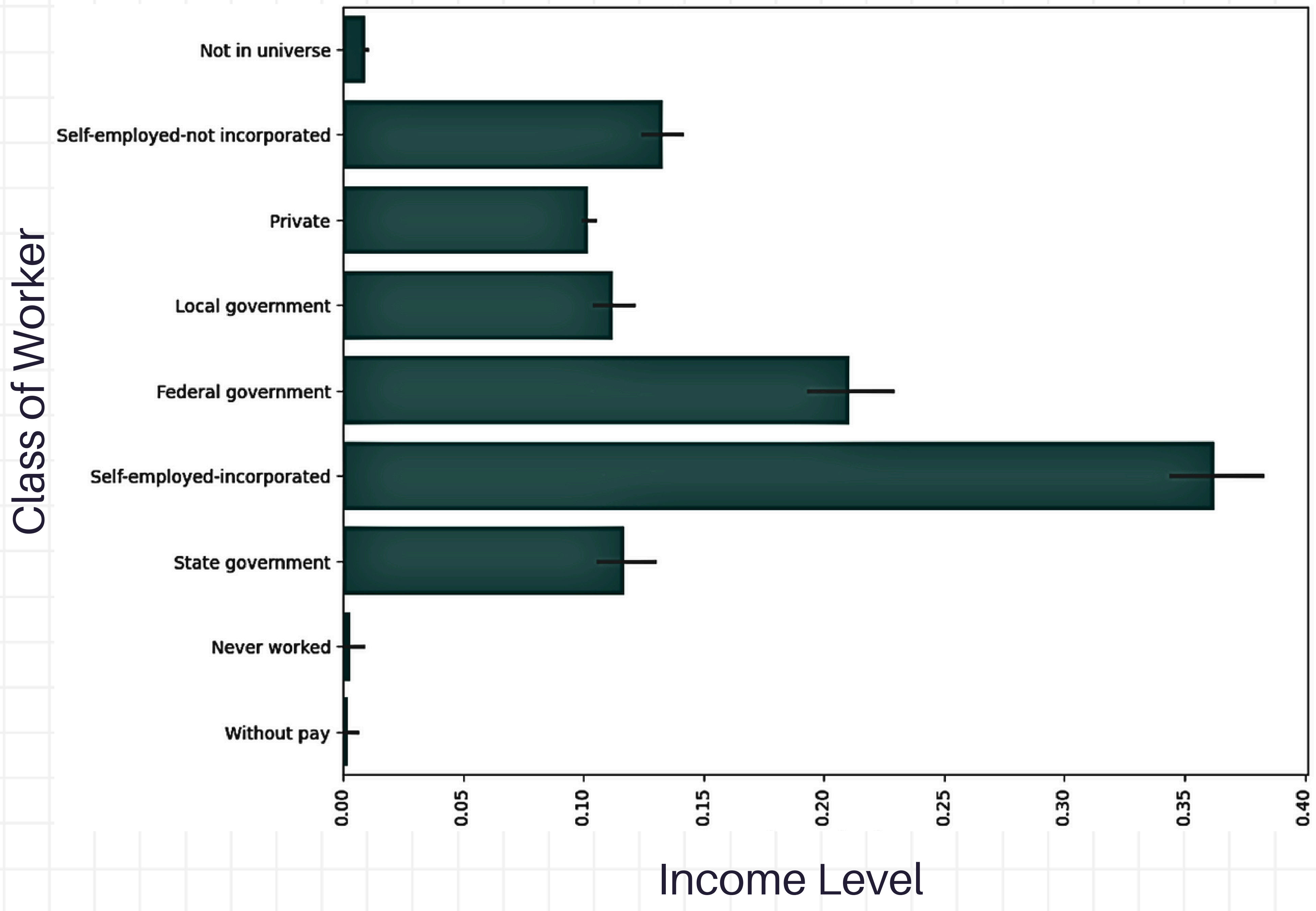
The United States Census Bureau that leads the country's Federal Statistical System

# Exploratory data analysis

- Class Imbalance
  - 187,141 <$50K
  - 12,382 >$50K
- Weighted% earning >$50K is 6.4%.

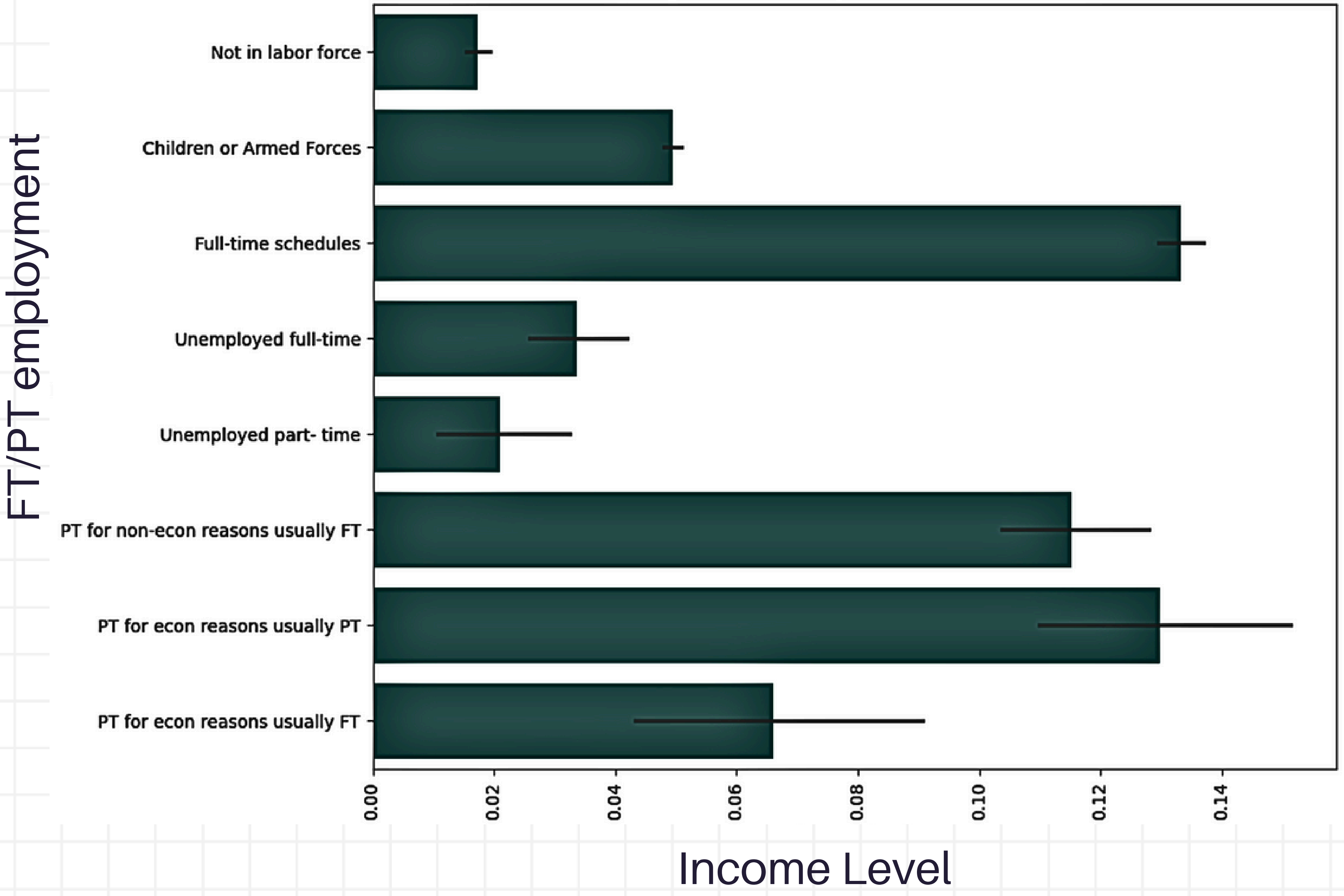


# Weighted% >$50K by Class of worker



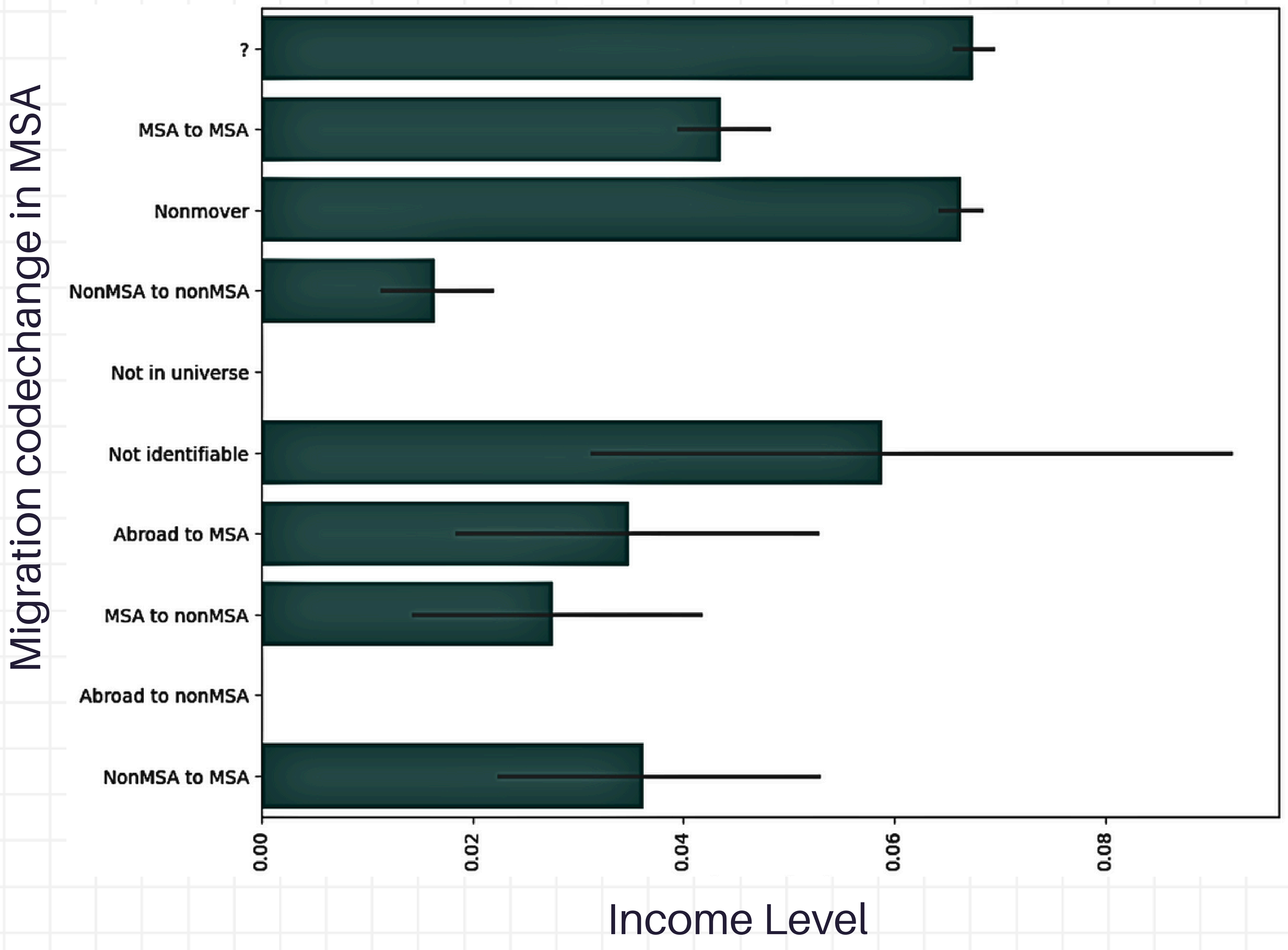

- **Self-employed (incorporated)** workers have the highest weighted% of earning >$50K at ~**35%.**
- **Federal government employees** have more weighted% of earning >$50K than state or local government at ~**20%.**
- Working in **state**, **local**, **private** or being **self-employed (not incorporated)** have moderate level of weighted% of earning >$50K at **12%, 12%, 10%** and **13%** respectively.

# Weighted% >$50K by Full-time(FT) or Part-time(PT) employment



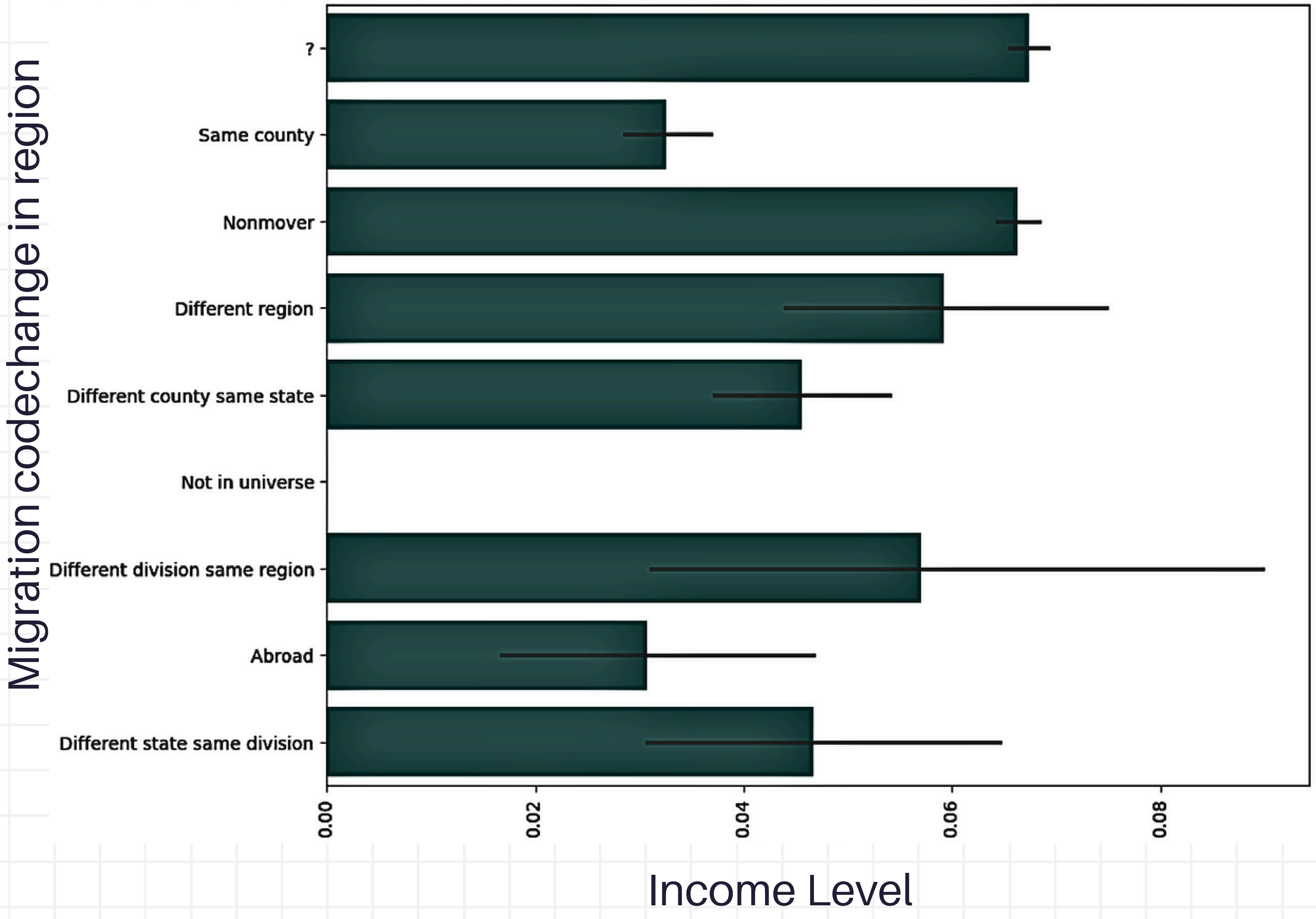

- **FT** earnings are strongly associated with earning >$50K (**~13%).**
- **PT for economic reasons and non-economic reasons** (usually PT or usually FT) also show elevated chances but are noticeably behind FT workers.
- **Non-labor groups** (~**1.5%**) and **unemployed groups** (~**2 - 3.5%**) rarely earn >$50K.

# Weighted% >$50K by Migration in Metropolitan statistical area (MSA)



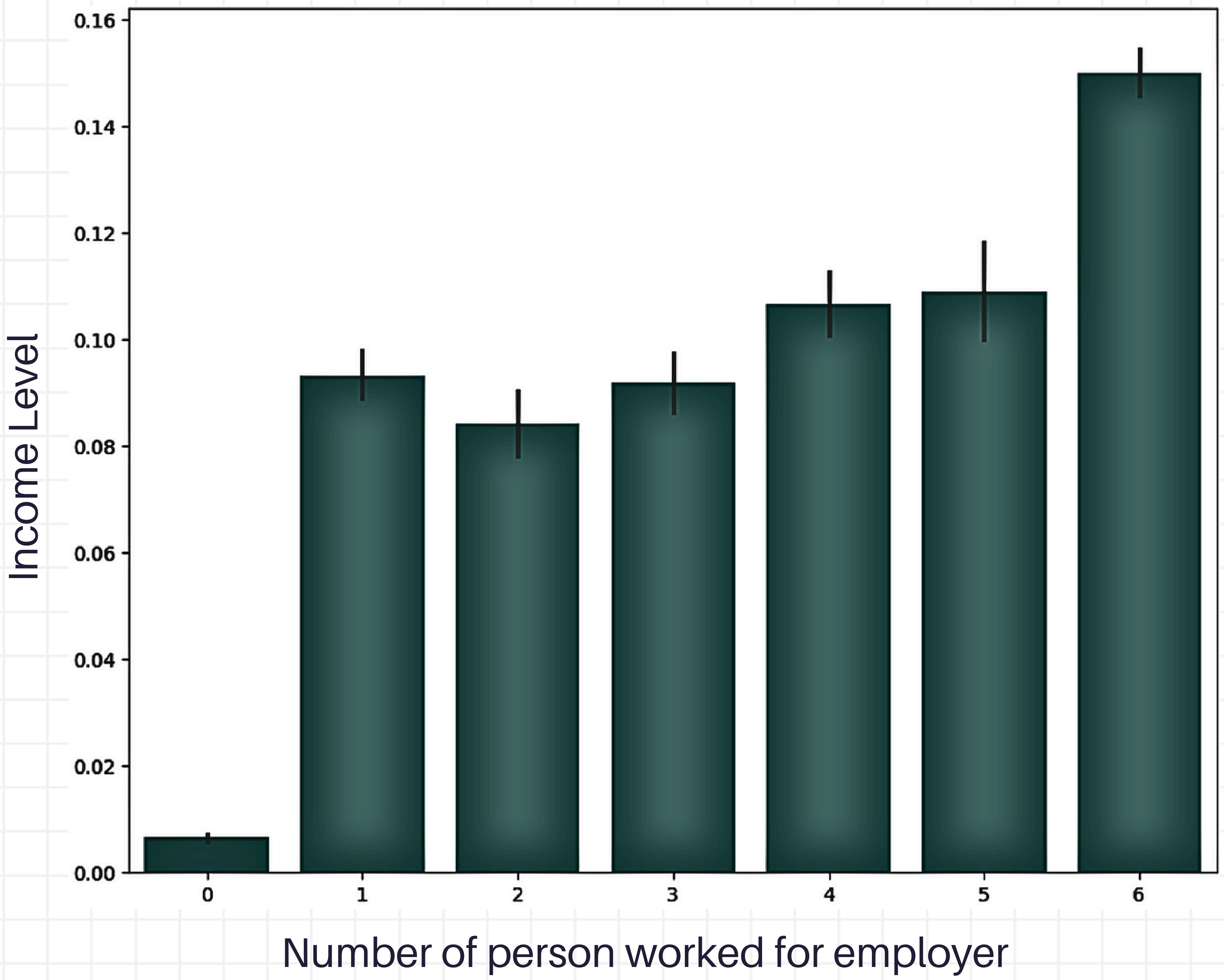

- **Non-movers** (~**6.5%**) or **intra-MSA movers** (~**4%**) are more likely to have >$50K earnings.
- Movers between **non-MSA to MSA** have lower high-income earnings of >$50K at ~**3.5%.**
- Movers between **MSA to non-MSA** have income earnings of >$50K at ~**2.5%.**

# Weighted% >$50K by Migration in Region



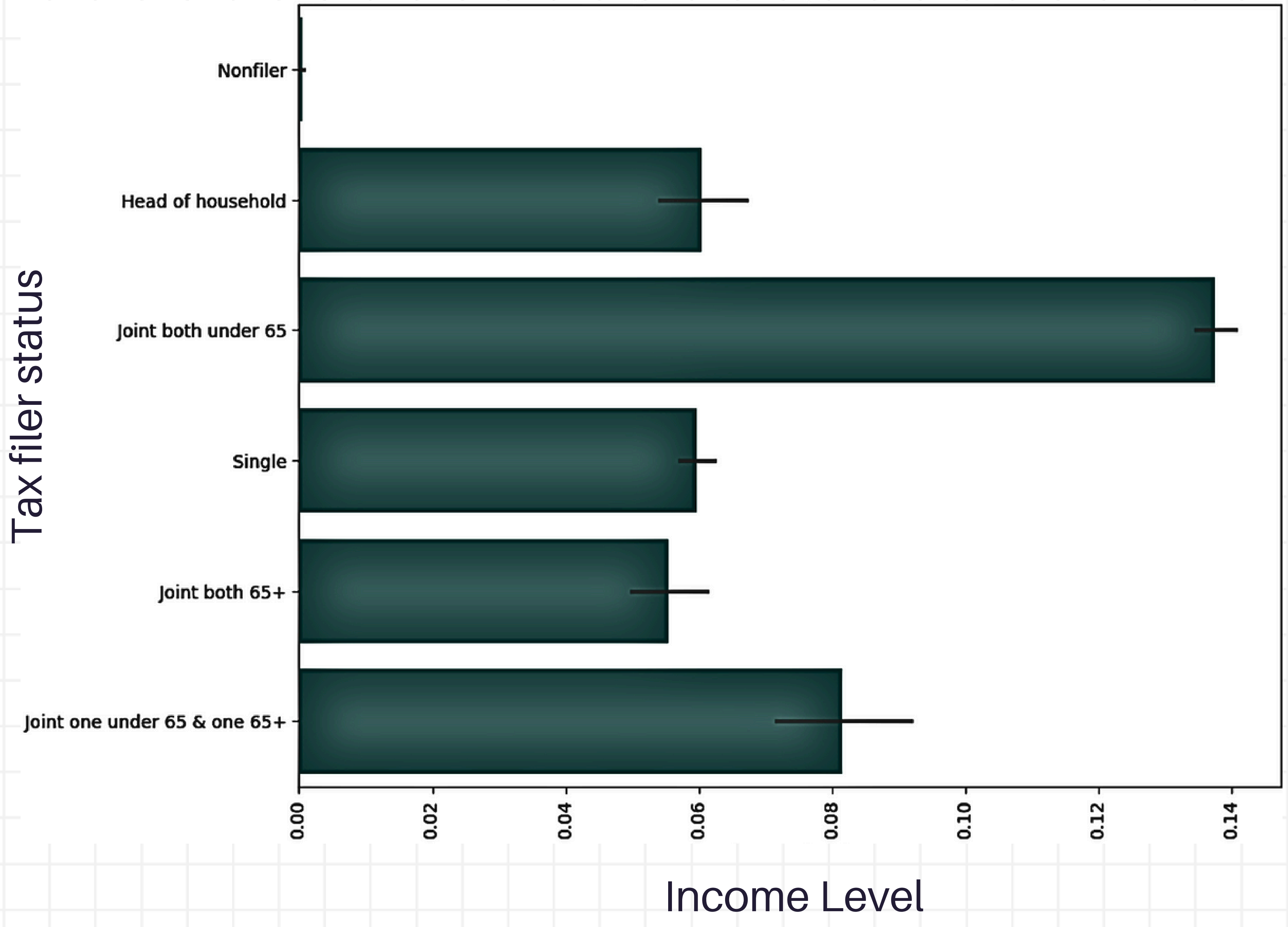

- **Non-movers** or individuals who stay in the same county have the highest weighted% earnings of >$50k at ~**6.5%.**
- **Migrating** across larger geographic areas (**region or division**) does not offer significant income advantage.
- Those who lived **abroad** or **who cannot be categorised due to data limitations,** exhibit lower, less predictable proportion of >$50K earners.

# Weighted% >$50K by Number of person worked for employer




- **Higher workplace size** correlates with **greater income**.
- **Sole workers** have the lowest likelihood of earning >$50K at **~0.5%.**
- Income advantage becomes most pronounced in **largest workplaces** (~**15%**).

# Weighted% >$50K by Tax filer status




- **Joint filers under 65** show the highest proportion of >$50K earners at ~**13.5%.**
- **Single filers** and **heads of household** are notably less likely to earn >$50K at ~ **5.5 - 6%.**
- **Non-filers** have an extremely low likelihood of high income.

# Feature engineering

- Balancing the classes (**Under sampling**)
- Grouping ages into range (**Bucketing**)
- Turning categories into easy-to-use numbers (**One-hot encoding**)
- Bringing numbers to a common scale (**Standardising** - **Weeks worked in a year**)

Plotted a correlation heatmap
for top 20% of features.
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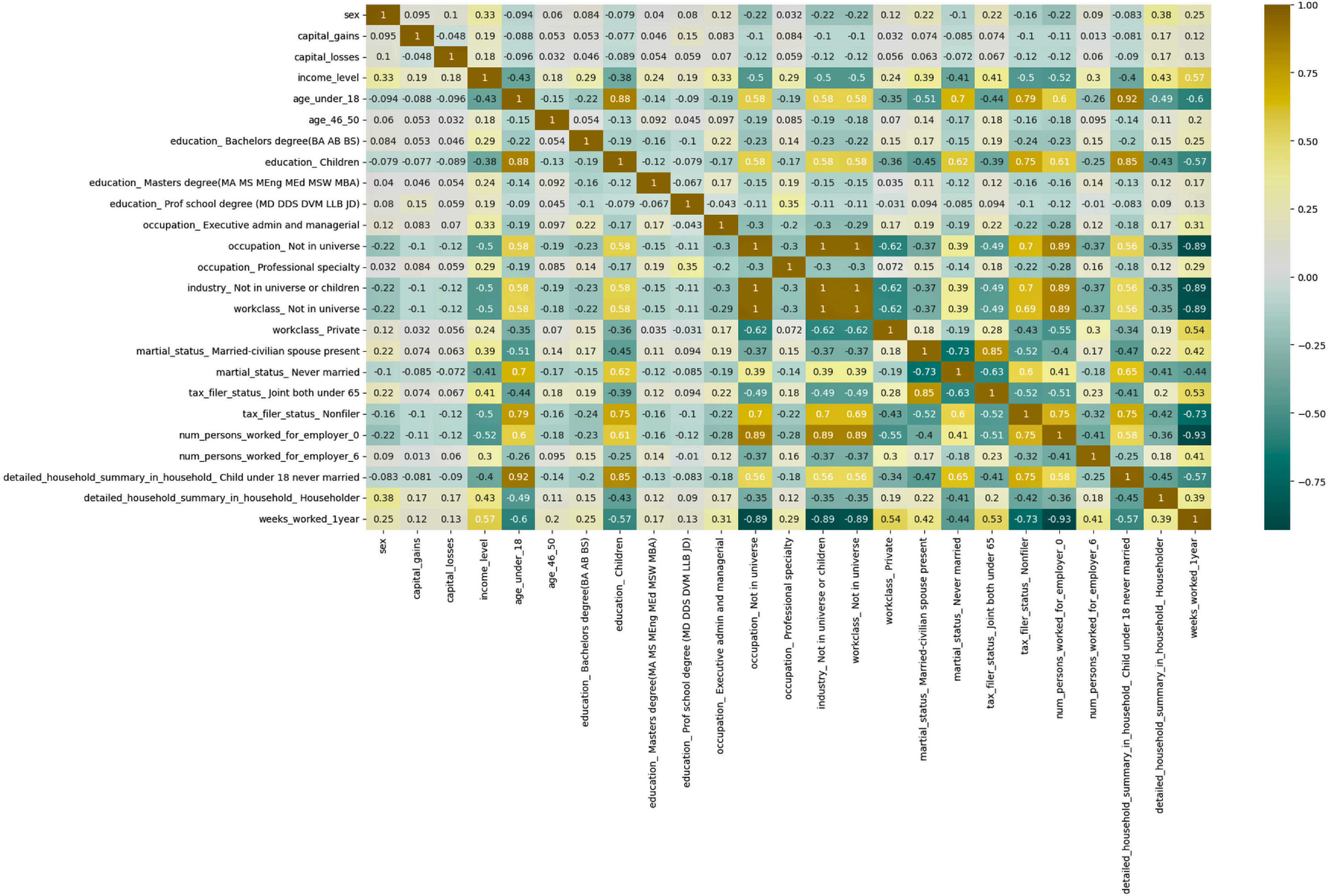
sex - 1 0.095 0.1 0.33 -0.094 0.06 0.084 -0.079 0.04 0.08 0.12 -0.22 0.032 -0.22 -0.22 0.12 0.22 -0.1 0.22 -0.16 -0.22 0.09 -0.083 0.38 0.25
capital_gains - 0.095 1 -0.048 0.19 -0.088 0.053 0.053 -0.077 0.046 0.15 0.083 -0.1 0.084 -0.1 -0.1 0.032 0.074 -0.085 0.074 -0.1 -0.11 0.013 -0.081 0.17 0.12
capital_losses - 0.1 -0.048 1 0.18 -0.096 0.032 0.046 -0.089 0.054 0.059 0.07 -0.12 0.059 -0.12 -0.12 0.056 0.063 -0.072 0.067 -0.12 -0.12 0.06 -0.09 0.17 0.13
income_level - 0.33 0.19 0.18 1 -0.43 0.18 0.29 -0.38 0.24 0.19 0.33 -0.5 0.29 -0.5 -0.5 0.24 0.39 -0.41 0.41 -0.5 -0.52 0.3 -0.4 0.43 0.57
age_under_18 - -0.094 -0.088 -0.096 -0.43 1 -0.15 -0.22 0.88 -0.14 -0.09 -0.19 0.58 -0.19 0.58 0.58 -0.35 -0.51 0.7 -0.44 0.79 0.6 -0.26 0.92 -0.49 -0.6
age_46_50 - 0.06 0.053 0.032 0.18 -0.15 1 0.054 -0.13 0.092 0.045 0.097 -0.19 0.085 -0.19 -0.18 0.07 0.14 -0.17 0.18 -0.16 -0.18 0.095 -0.14 0.11 0.2
education_ Bachelors degree(BA AB BS) - 0.084 0.053 0.046 0.29 -0.22 0.054 1 -0.19 -0.16 -0.1 0.22 -0.23 0.14 -0.23 -0.22 0.15 0.17 -0.15 0.19 -0.24 -0.23 0.15 -0.2 0.15 0.25
education_ Children - -0.079 -0.077 -0.089 -0.38 0.88 -0.13 -0.19 1 -0.12 -0.079 -0.17 0.58 -0.17 0.58 0.58 -0.36 -0.45 0.62 -0.39 0.75 0.61 -0.25 0.85 -0.43 -0.57
education_ Masters degree(MA MS MEng MEd MSW MBA) - 0.04 0.046 0.054 0.24 -0.14 0.092 -0.16 -0.12 1 -0.067 0.17 -0.15 0.19 -0.15 -0.15 0.035 0.11 -0.12 0.12 -0.16 -0.16 0.14 -0.13 0.12 0.17
education_ Prof school degree (MD DDS DVM LLB JD) - 0.08 0.15 0.059 0.19 -0.09 0.045 -0.1 -0.079 -0.067 1 -0.043 -0.11 0.35 -0.11 -0.11 -0.031 0.094 -0.085 0.094 -0.1 -0.12 -0.01 -0.083 0.09 0.13
occupation_ Executive admin and managerial - 0.12 0.083 0.07 0.33 -0.19 0.097 0.22 -0.17 0.17 -0.043 1 -0.3 -0.2 -0.3 -0.29 0.17 0.19 -0.19 0.22 -0.22 -0.28 0.12 -0.18 0.17 0.31
occupation_ Not in universe - -0.22 -0.1 -0.12 -0.5 0.58 -0.19 -0.23 0.58 -0.15 -0.11 -0.3 1 -0.3 1 1 -0.62 -0.37 0.39 -0.49 0.7 0.89 -0.37 0.56 -0.35 -0.89
occupation_ Professional specialty - 0.032 0.084 0.059 0.29 -0.19 0.085 0.14 -0.17 0.19 0.35 -0.2 -0.3 1 -0.3 -0.3 0.072 0.15 -0.14 0.18 -0.22 -0.28 0.16 -0.18 0.12 0.29
industry_ Not in universe or children - -0.22 -0.1 -0.12 -0.5 0.58 -0.19 -0.23 0.58 -0.15 -0.11 -0.3 1 -0.3 1 1 -0.62 -0.37 0.39 -0.49 0.7 0.89 -0.37 0.56 -0.35 -0.89
workclass_ Not in universe - -0.22 -0.1 -0.12 -0.5 0.58 -0.18 -0.22 0.58 -0.15 -0.11 -0.29 1 -0.3 1 1 -0.62 -0.37 0.39 -0.49 0.69 0.89 -0.37 0.56 -0.35 -0.89
workclass_ Private - 0.12 0.032 0.056 0.24 -0.35 0.07 0.15 -0.36 0.035 -0.031 0.17 -0.62 0.072 -0.62 -0.62 1 0.18 -0.19 0.28 -0.43 -0.55 0.3 -0.34 0.19 0.54
martial_status_ Married-civilian spouse present - 0.22 0.074 0.063 0.39 -0.51 0.14 0.17 -0.45 0.11 0.094 0.19 -0.37 0.15 -0.37 -0.37 0.18 1 -0.73 0.85 -0.52 -0.4 0.17 -0.47 0.22 0.42
martial_status_ Never married - -0.1 -0.085 -0.072 -0.41 0.7 -0.17 -0.15 0.62 -0.12 -0.085 -0.19 0.39 -0.14 0.39 0.39 -0.19 -0.73 1 -0.63 0.6 0.41 -0.18 0.65 -0.41 -0.44
tax_filer_status_ Joint both under 65 - 0.22 0.074 0.067 0.41 -0.44 0.18 0.19 -0.39 0.12 0.094 0.22 -0.49 0.18 -0.49 -0.49 0.28 0.85 -0.63 1 -0.52 -0.51 0.23 -0.41 0.2 0.53
tax_filer_status_ Nonfiler - -0.16 -0.1 -0.12 -0.5 0.79 -0.16 -0.24 0.75 -0.16 -0.1 -0.22 0.7 -0.22 0.7 0.69 -0.43 -0.52 0.6 -0.52 1 0.75 -0.32 0.75 -0.42 -0.73
num_persons_worked_for_employer_0 - -0.22 -0.11 -0.12 -0.52 0.6 -0.18 -0.23 0.61 -0.16 -0.12 -0.28 0.89 -0.28 0.89 0.89 -0.55 -0.4 0.41 -0.51 0.75 1 -0.41 0.58 -0.36 -0.93
num_persons_worked_for_employer_6 - 0.09 0.013 0.06 0.3 -0.26 0.095 0.15 -0.25 0.14 -0.01 0.12 -0.37 0.16 -0.37 -0.37 0.3 0.17 -0.18 0.23 -0.32 -0.41 1 -0.25 0.18 0.41
detailed_household_summary_in_household_ Child under 18 never married - -0.083 -0.081 -0.09 -0.4 0.92 -0.14 -0.2 0.85 -0.13 -0.083 -0.18 0.56 -0.18 0.56 0.56 -0.34 -0.47 0.65 -0.41 0.75 0.58 -0.25 1 -0.45 -0.57
detailed_household_summary_in_household_ Householder - 0.38 0.17 0.17 0.43 -0.49 0.11 0.15 -0.43 0.12 0.09 0.17 -0.35 0.12 -0.35 -0.35 0.19 0.22 -0.41 0.2 -0.42 -0.36 0.18 -0.45 1 0.39
weeks_worked_1year - 0.25 0.12 0.13 0.57 -0.6 0.2 0.25 -0.57 0.17 0.13 0.31 -0.89 0.29 -0.89 -0.89 0.54 0.42 -0.44 0.53 -0.73 -0.93 0.41 -0.57 0.39 1
sex
capital_gains
capital_losses
income_level
age_under_18
age_46_50
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workclass_ Private
martial_status_ Married-civilian spouse present
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num_persons_worked_for_employer_0
num_persons_worked_for_employer_6
detailed_household_summary_in_household_ Child under 18 never married
detailed_household_summary_in_household_ Householder
weeks_worked_1year
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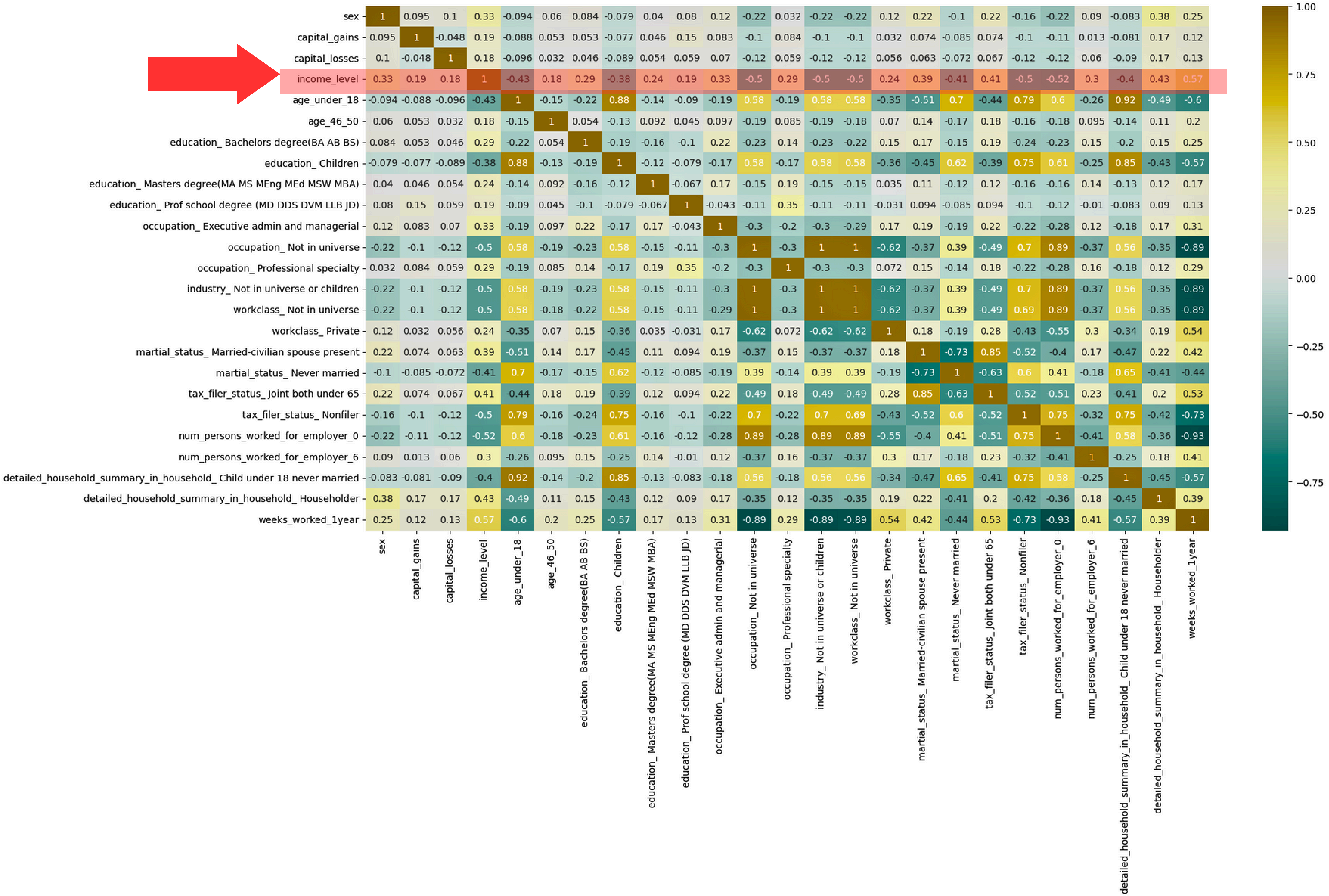
0.25
0.00
−0.25
−0.50
−0.75



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weeks_worked_1year 0.25 0.12 0.13 0.57 -0.6 0.2 0.25 -0.57 0.17 0.13 0.31 -0.89 0.29 -0.89 -0.89 0.54 0.42 -0.44 0.53 -0.73 -0.93 0.41 -0.57 0.39 1
sex
capital_gains
capital_losses
income_level
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age_46_50
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education_ Prof school degree (MD DDS DVM LLB JD)
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occupation_ Not in universe
occupation_ Professional specialty
industry_ Not in universe or children
workclass_ Not in universe
workclass_ Private
martial_status_ Married-civilian spouse present
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tax_filer_status_ Joint both under 65
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num_persons_worked_for_employer_6
detailed_household_summary_in_household_ Child under 18 never married
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weeks_worked_1year
1.00
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0.50
0.25
0.00
−0.25
−0.50
−0.75

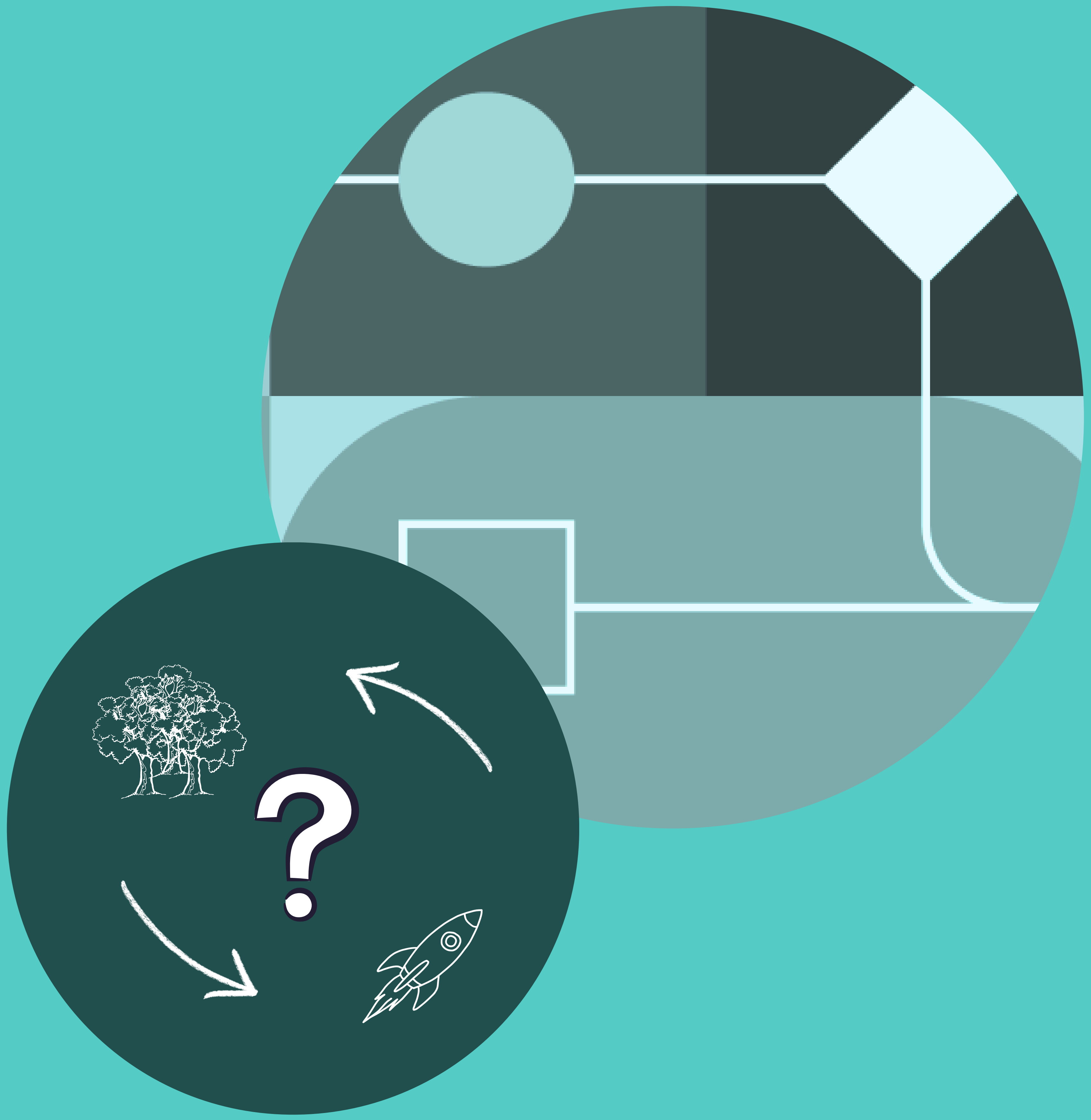

**Features most strongly linked to higher income:** Weeks worked, employer size, higher education, occupation status, capital gains, being male, filing taxes jointly

**Features most strongly linked to lower income:** Underage status, not engaged in workforce, nonfiler tax status, presence of children, never married

# Modeling and evaluation

- **Random forest:** Combines results of many decision trees to make answers more reliable and less prone to mistakes.

- **XGBoost:** Builds knowledge step by step, steadily improving by focusing on those the model got wrong the first time.



# Random Forest Classifier (RFC)

### Precision

- <$50K = 89%
- >$50K = 85%

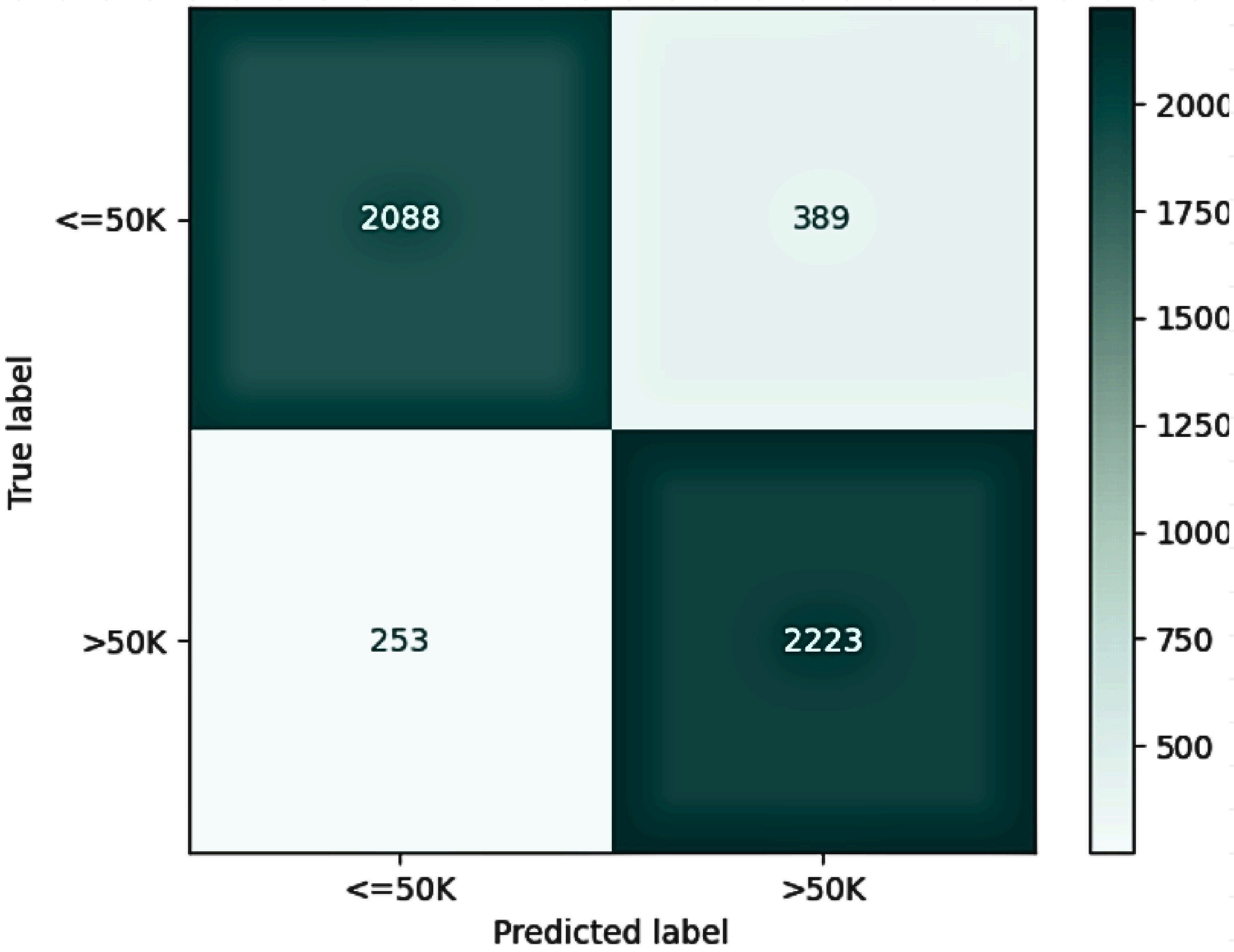
### Recall

- <$50K = 84%
- >$50K = 90%

### Overall Accuracy

87%

Confusion Matrix-RFC



# Feature Importances (RFC)

| | feature | importance |
|---|---|---|
| 0 | dividends_from_stocks | 0.075028 |
| 1 | weeks_worked_1year | 0.074025 |
| 2 | num_persons_worked_for_employer_0 | 0.049138 |
| 3 | detailed_household_summary_in_household_ House... | 0.042650 |
| 4 | sex | 0.039658 |
| 5 | capital_gains | 0.035647 |
| 6 | industry_ Not in universe or children | 0.029186 |
| 7 | tax_filer_status_ Nonfiler | 0.025461 |
| 8 | occupation_ Not in universe | 0.024434 |
| 9 | occupation_ Executive admin and managerial | 0.023406 |
| 10 | martial_status_ Never married | 0.023347 |
| 11 | tax_filer_status_ Joint both under 65 | 0.022605 |
| 12 | workclass_ Not in universe | 0.022335 |
| 13 | education_ Bachelors degree(BA AB BS) | 0.020697 |
| 14 | occupation_ Professional specialty | 0.019770 |

- The most influential factors, according to the RFC model, are shown here in order of importance.
- Features like **stock dividends, weeks worked in a year, and workplace size** have the greatest impact on predicting whether someone earns above $50K.
- **Household details, gender, and financial variables** (e.g. **capital gains**) also play key roles.
- The model uses this information to find the patterns most associated with higher income.
- Can interpret model's recommendations and direct attention to factors that could be addressed by policy or business action.

# XGBoost

**Precision**

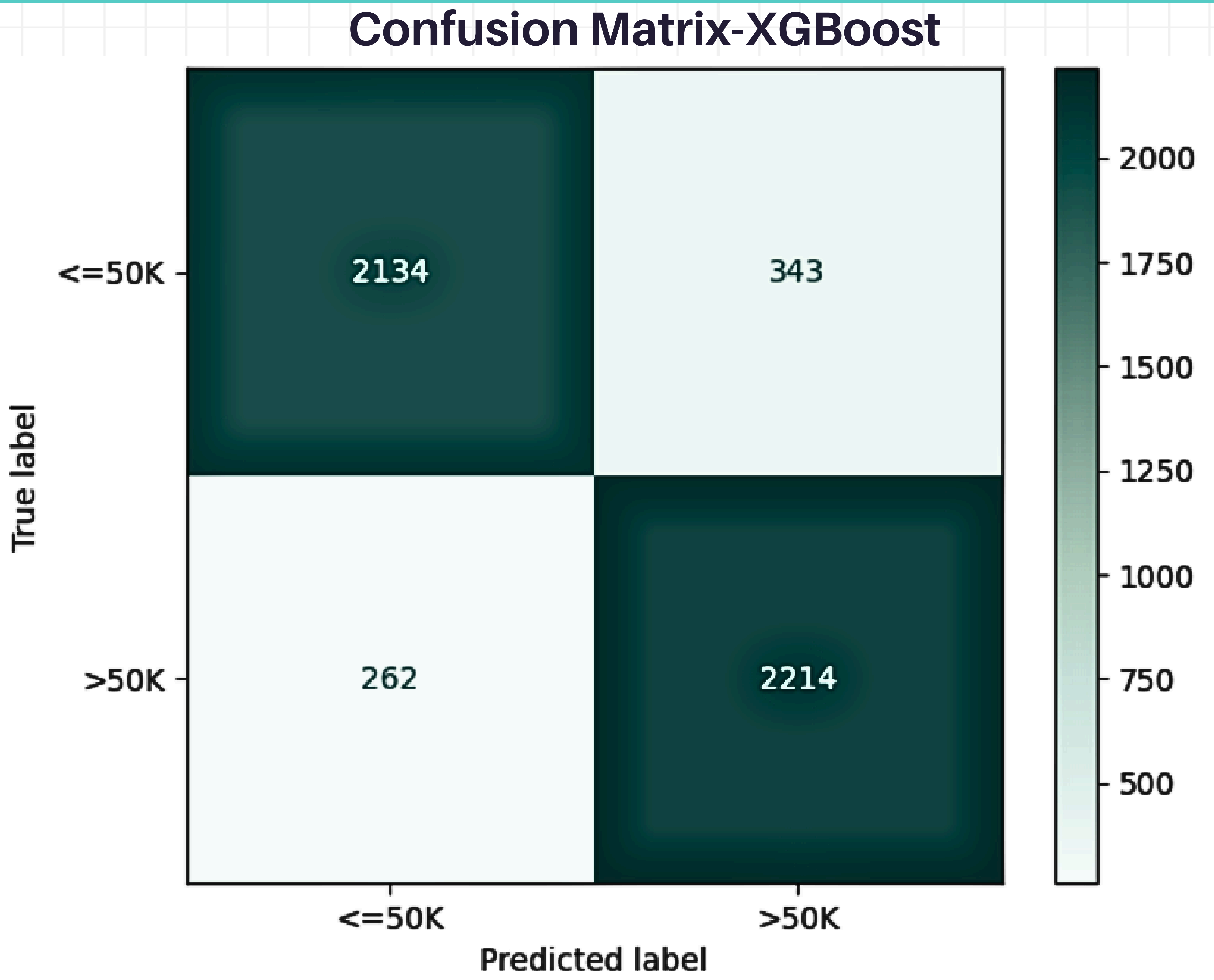
- <$50K = 89%
- >$50K = 87%

**Recall**

- <$50K = 86%
- >$50K = 89%

**Overall Accuracy**

88%



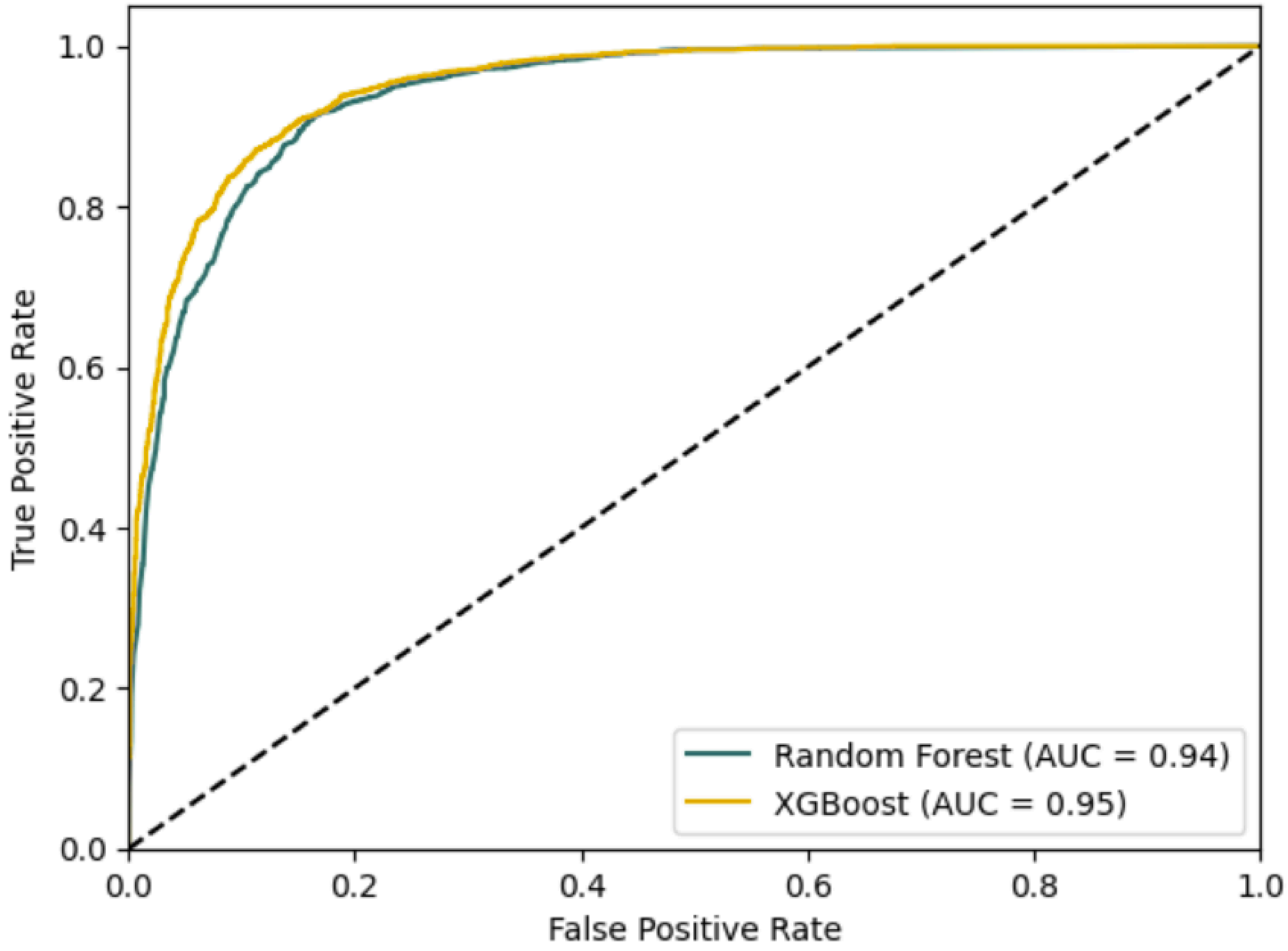
# Feature Importances (XGBoost)

| | feature | importance |
|---|---|---|
| 0 | tax_filer_status_ Nonfiler | 0.130113 |
| 1 | weeks_worked_1year | 0.089042 |
| 2 | occupation_ Other service | 0.056085 |
| 3 | age_under_18 | 0.039131 |
| 4 | occupation_ Executive admin and managerial | 0.026587 |
| 5 | sex | 0.025874 |
| 6 | occupation_ Professional specialty | 0.024198 |
| 7 | dividends_from_stocks | 0.021031 |
| 8 | education_ Prof school degree (MD DDS DVM LLB JD) | 0.020137 |
| 9 | martial_status_ Never married | 0.019911 |
| 10 | education_ Masters degree(MA MS MEng MEd MSW MBA) | 0.019136 |
| 11 | age_21_25 | 0.018913 |
| 12 | capital_gains | 0.018840 |
| 13 | occupation_ Handlers equip cleaners etc | 0.017261 |
| 14 | occupation_ Adm support including clerical | 0.015607 |

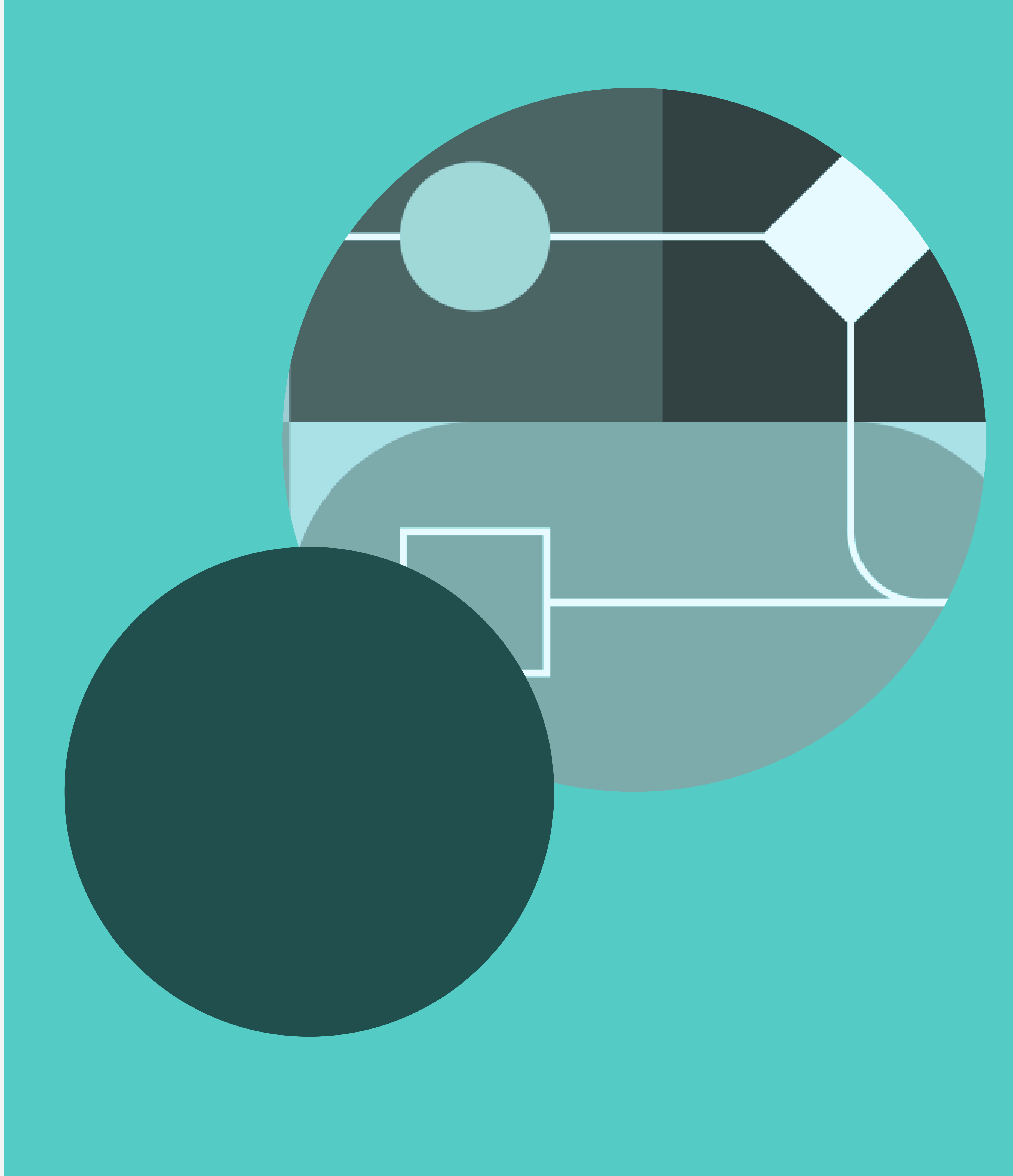
- The most influential factors, according to the XGBoost model, are shown here in order of importance.
- **Tax filer status** (**non-filer**) **and weeks worked in a year** are most powerful indicators, meaning these features played the biggest roles in the model's decision making.
- **Job types, certain age groups, and educational attainment** also significantly influenced predictions.
- Can interpret model's recommendations and direct attention to factors that could be addressed by policy or business action.

# ROC-AUC Comparison

- Receiver operating characteristics
- Area under the curve
- The higher the curve the better the model



# Results



# Predicting on Final Dataset

### Precision

- <$50K = 99%
- >$50K = 30%

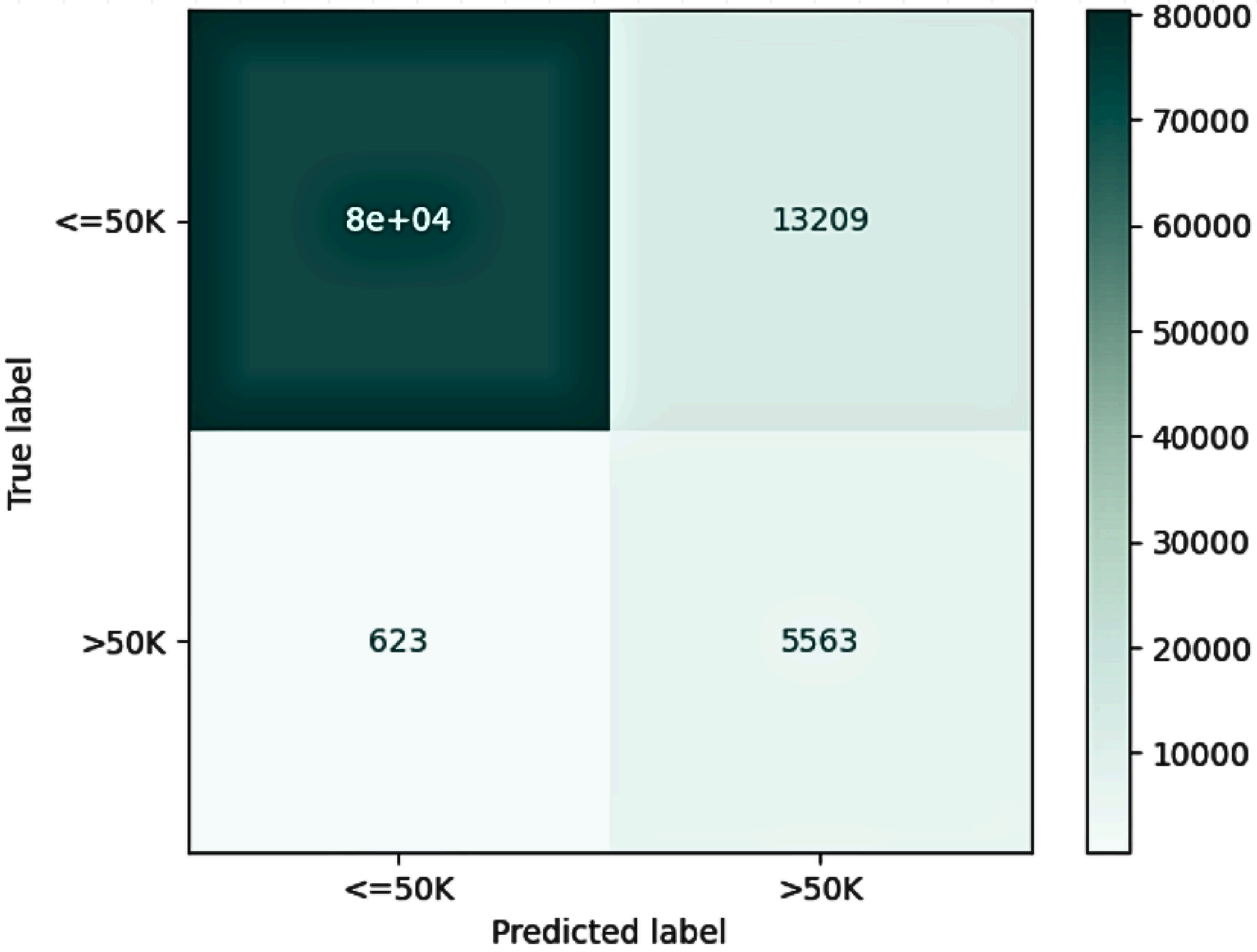
### Recall

- <$50K = 86%
- >$50K = 90%

### Overall Accuracy

86%

## Confusion Matrix-XGBoost



# Recommendations

## Marriage

- Promote marriages
- Stable households
- Joint tax filing
- Community building and support programs

## Education

- Promote advanced education
- Boost individuals earning potential

## Career

- Fostering career mobility within large organisations
- Encouraging entrepreneurship

## Financial awareness

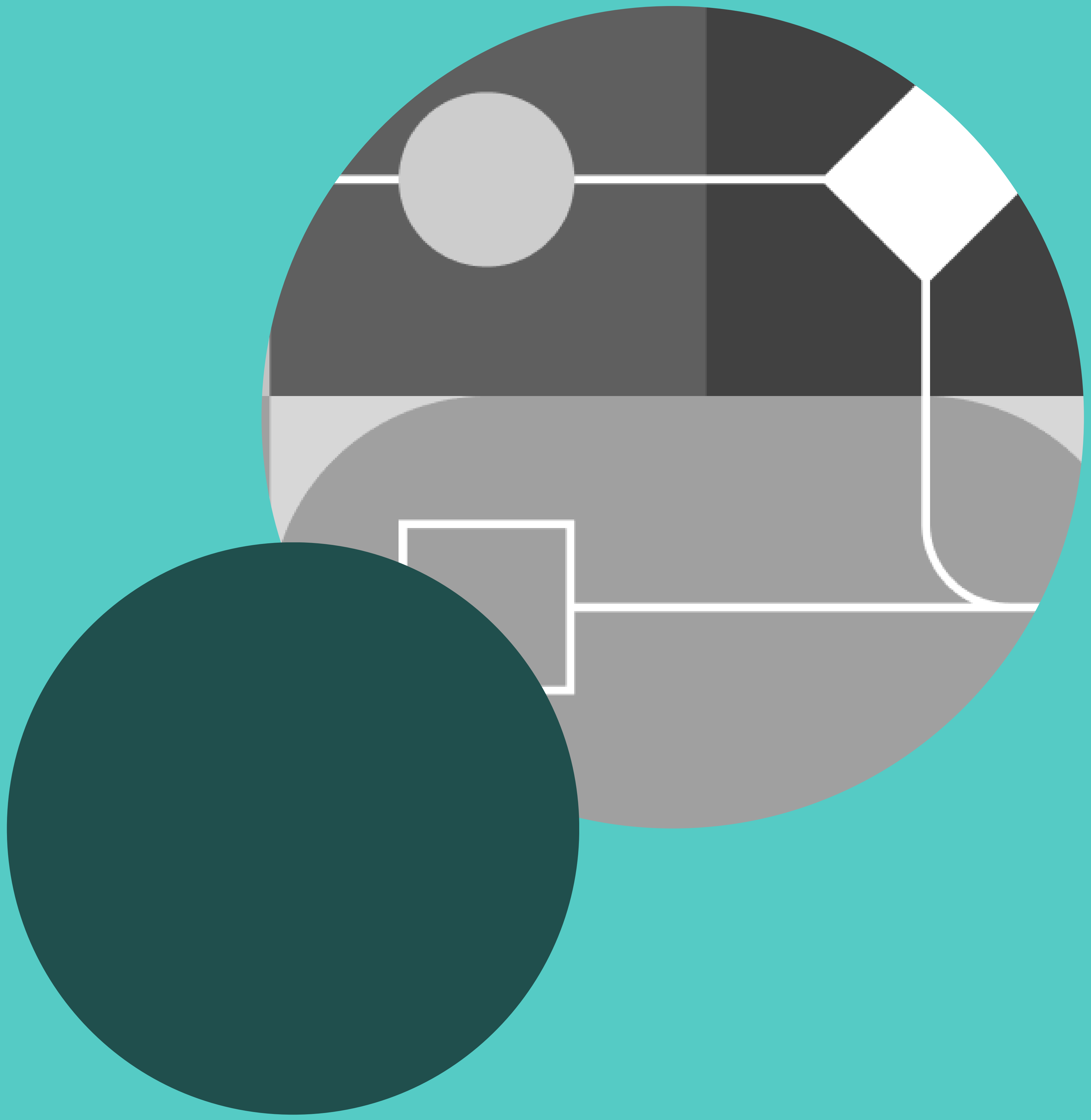
- Promoting financial awareness
- Investment awareness in stocks, dividends, etc.
- Improving long term outcomes

# Limitations

- Dataset based on stratified sampling
- Feature coverage and data quality
- Data doesn’t reflect current or future income patterns due to dynamic market
- Biasness in data due to sampling methods

# Future work

- Model enhancement
- Feature engineering
- Over sampling methods
- Feedback loop



# Thank you!